

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

COMPUTER SCIENCE

Examination, May 2021

Nature-Inspired Computation

Module Leader: Professor Edward Keedwell

Duration: TWO HOURS + 30 MINUTES UPLOAD TIME

Answer ALL questions.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

This is an OPEN BOOK examination.

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Question 1

- (a) The following describes three representations that might be used within an evolutionary algorithm:

- (1) Permutation representation using the letters from A to L
- (2) Binary representation of length 12
- (3) K-Ary representation of length 12 and a K of 8

- (i) Provide an example of a single random solution for each of the representation types.

(6 marks)

- (ii) Name and briefly describe the most appropriate mutation operators for each representation.

(6 marks)

- (iii) Which representation would be most useful for the following problems? Explain your answer.

- A) The knapsack problem, where decisions must be made as to which items to pack into a knapsack, maximising value within weight constraints
- B) The travelling salesman problem, where a route must be devised that visits all cities once and doesn't revisit any city
- C) Water distribution network optimisation, where a set of pipe diameters are selected from a discrete set

(6 marks)

- (iv) Is standard single-point crossover likely to work well for all these representations? Explain your answer.

(4 marks)

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- (b) Nature-inspired methods attempt to balance *exploration* against *exploitation* during the search process. This balance can often be altered through the modification of the algorithm's parameters. For each of the parameters shown below, write down a value that would promote *exploration* and a value that would promote *exploitation* and explain why these values produce these effects.

(i) Tournament size (with a population size of 100).

(4 marks)

(ii) Population size for an evolutionary algorithm.

(4 marks)

(iii) c_2 .

(4 marks)

- (c) Multi-layer perceptrons and self-organising maps are both methods inspired by the brain. Briefly explain three main differences between these two neural techniques.

(6 marks)

(Total 40 marks)

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Question 2

The University has asked you to develop a new lecture timetabling solution based on a nature-inspired algorithm of your choice. The algorithm should allocate lectures for a set of modules to a finite set of lecture rooms with varying capacity within the University. The number of students on each module will be provided to you, along with the capacity of each room and the staff members required to teach and their availability. Students would prefer not to have lectures at 8.30am and 5.30pm.

- (a) Describe, and appropriately label, the hard and soft constraints of this problem.

(5 marks)

- (b) Describe how you would formulate the objective function(s) for this problem.

(5 marks)

- (c) Describe and justify your choice of algorithm to solve this problem and describe the representation you would use.

(8 marks)

- (d) Describe the parameter settings you would use and what experiments you would conduct to determine the efficacy of your choices.

(7 marks)

- (e) Explain what ‘success’ would look like on completion of this project.

(5 marks)

(Total 30 marks)

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Question 3

You have been given a classification problem to solve with a perceptron with an activation function, φ that takes the value 1 when the input is greater than or equal to 0, and 0 otherwise. The training set for the problem is defined as follows:

Input (x_1)	Input (x_2)	Target (y)
0	3	0
-1	2	0
-1	0	1
1	2	1

- (a) Sketch a decision boundary that separates the two different classes in y and specify its equation.

(6 marks)

- (b) Design a single-neuron perceptron to solve this classification problem by drawing a diagram and writing down a possible solution of weights and biases that would lead to the behaviour above.

(8 marks)

- (c) Test your solution with all four input patterns, writing down your calculations.

(8 marks)

- (d) Describe the changes that would be necessary to convert this perceptron into a multi-layer perceptron (MLP).

(4 marks)

- (e) Describe the main steps of training the MLP on training data.

(4 marks)

(Total 30 marks)